Highlights

UAV-TAlign-12K: Scene-Level Reliability Evaluation for Cross-FOV UAV Infrared-Visible Image Alignment

Anonymous Authors

- UAV-TAlign-12K provides a 6,039-pair UAV infrared-visible alignment collection.
- A label-free scene-level reliability protocol evaluates retained alignment products.
- RIFT2 and modern dense baselines show broadly available pairwise homography evidence.
- Robust consensus and QA-aware verification convert pairwise evidence into retained transforms.

UAV-TAlign-12K: Scene-Level Reliability Evaluation for Cross-FOV UAV Infrared-Visible Image Alignment

Anonymous Authors[a]

[a] *Anonymous submission,*

Abstract

UAV infrared-visible alignment is a prerequisite for reliable aerial inspection, nighttime monitoring, and multimodal thermal sensing. In cross-FOV aerial imagery, the operating question is not only whether two frames can be matched, but whether the resulting alignment should be retained as a scene-level product. We introduce UAV-TAlign-12K, a 6,039-pair UAV infrared-visible alignment collection with an integrity-checked 6,037-pair evaluation split, and a scalable label-free protocol for reliability-controlled scene evaluation. We further present UAV-TAlign, a scene-level alignment framework that converts strong pairwise multimodal evidence into retained scene transforms through reliability-guided evidence selection, robust scene consensus, and QA-aware verification. Journal-scale experiments show that pairwise homography evidence is broadly available: RIFT2, raw MINIMA, and RoMa produce homographies for all 6,037 evaluated pairs, while LoFTR and XoFTR remain near-saturated. Under the strict canonical QA gate, UAV-TAlign produces scene-level homographies for all 15 scenes and retains 9 high-confidence alignment products. The risk-coverage profile further shows how reliability-controlled retention can be studied as an operating trade-off rather than a single pairwise matching score.

Keywords: infrared-visible registration, UAV imagery, thermal imaging, multimodal alignment, scene-level reliability

1. Introduction

Infrared-visible alignment is becoming a foundational capability for UAV inspection, monitoring, and nighttime multimodal perception. Visible imagery provides semantic and structural detail, while infrared thermal imagery remains informative under illumination changes, smoke, haze, and low-light conditions. This complementarity has driven progress across multispectral pedestrian perception, thermal object detection, visible-thermal tracking, and UAV multimodal sensing [1, 2, 3, 4, 5, 6, 7, 8]. For these systems to operate as fused sensors rather than separate streams, however, RGB and thermal observations must be aligned reliably at the scene level.

Real UAV infrared-visible alignment is substantially more demanding than standard same-view image matching. Aerial platforms introduce viewpoint drift, cross-FOV capture, rolling scale changes, and cross-resolution sensors; thermal images further introduce low-texture regions, grayscale or pseudocolor rendering, and strong day/night appearance shifts. These factors make individual frame-level homographies uneven: some pairs are strongly supported, while others require additional scene context before they should influence the final transform. The practical question is therefore not only whether a matcher can produce correspondences for one pair, but whether a system can decide which scene-level alignment should be trusted.

Recent pairwise matching has advanced rapidly. Transformer, dense, and multimodal matchers such as LoFTR, LightGlue, DKM, RoMa, XoFTR, and MINIMA have substantially improved correspondence availability under difficult appearance changes [9, 10, 11, 12, 13, 14, 15, 16]. At the benchmark level, recent UAV visible-infrared alignment studies have also shown that synthetic transform supervision can support trainable alignment baselines with controlled geometric metrics [17]. UAV-TAlign-12K targets the complementary operating regime: real captured cross-FOV UAV infrared-visible scenes where the central product is a retained scene-level transform rather than only a supervised pairwise transform prediction. Our 12K-scale experiments confirm this progress in the UAV infrared-visible setting: strong off-the-shelf multimodal matchers and an infrared-visible classical baseline already provide broadly available pairwise homography evidence on UAV-TAlign-12K. This result creates a concrete tension: a method can approach saturated pairwise usability while producing only selective scene-level retention under a strict acceptance rule. Pairwise usability and scene-level canonical acceptance therefore measure different operating questions and should not be read as the same success criterion. The key step is to aggregate evidence across a scene, suppress unstable homographies, and verify whether the final transform should be retained.

This observation motivates a scene-level framing of the problem. We study UAV infrared-visible registration as a reliability-centered alignment task rather than a collection of independent pairwise matches. Under this view, modern matchers are powerful evidence generators, but the alignment algorithm must still answer three system-level questions: which frames should contribute evidence, how should multiple homographies be fused into a scene consensus, and when is the resulting transform reliable enough to use? This framing is especially important for UAV data, where the final output should favor trustworthy scene-level geometry over indiscriminate acceptance.

We present UAV-TAlign, a scene-level UAV infrared-visible alignment framework designed around this decision layer. UAV-TAlign operates on top of modern pairwise matchers and uses reliability-guided evidence selection to obtain representative frame support, robust scene consensus to consolidate frame-level estimates, and QA-aware verification to retain scene-level transforms that satisfy strict quality criteria. In this sense, UAV-TAlign elevates strong pairwise multimodal matching into a deployable scene-level alignment decision.

We also package UAV-TAlign-12K, a public UAV infrared-visible alignment collection with 6,039 candidate pairs and an integrity-checked 6,037-pair evaluation split across 15 scenes. The collection spans day, night, low-light, grayscale thermal, pseudocolor thermal, and wide/zoom scene conditions. It is accompanied by a scalable label-free protocol that evaluates pairwise homography evidence, scene-level retention, reliability diagnostics, and risk-coverage operating profiles. Together, the data and protocol provide a practical and reproducible way to study scene-level alignment reliability under real UAV infrared-visible conditions.

Our contributions are threefold:

- We introduce UAV-TAlign, a scene-level UAV infrared-visible alignment framework with reliability-guided evidence selection, robust scene consensus, and QA-aware verification.

- We package UAV-TAlign-12K, a 6,039-pair cross-FOV UAV infrared-visible alignment collection with an integrity-checked 6,037-pair evaluation split and a scalable label-free reliability protocol.

- We show through infrared-visible classical and modern dense baselines, reliability diagnostics, and cumulative analyses that the central operating challenge shifts from pairwise match availability to reliability-controlled scene retention.

2. Related Work

2.1. Classical Image Registration

Classical image registration pipelines typically rely on hand-crafted local features followed by robust geometric fitting. Representative methods built on SIFT, SURF, ORB, BRISK, KAZE, or AKAZE remain attractive because they are simple, interpretable, and do not require task-specific training [18, 19, 20, 21, 22, 23]. In cross-modal settings, the appearance gap between visible and thermal imagery can reduce descriptor repeatability, especially when thermal frames contain low-texture or homogeneous regions. These methods nevertheless provide a strong geometric reference point for understanding which parts of the registration problem are addressed by local feature extraction and which parts require higher-level scene reasoning.

These methods remain important baselines in our study because they represent the classical registration view of the problem: stable local features followed by robust homography fitting. Robust model estimation remains central in these pipelines, from RANSAC and multiple-view geometry to more recent variants such as MAGSAC and MAGSAC++ [24, 25, 26, 27]. Our results show that real UAV infrared-visible scenes benefit from moving beyond a single-pair formulation toward scene-level evidence aggregation and reliability control.

2.2. Deep Pairwise Matching for Multimodal Registration

Recent dense and semi-dense matching methods have significantly improved pairwise geometric correspondence estimation. Learned local features have evolved from trainable detectors and descriptors such as SuperPoint, LF-Net, D2-Net, R2D2, DISK, S2DNet, HardNet, L2-Net, SOSNet, GeoDesc, ContextDesc, ASLFeat, ALIKE, and Key.Net [28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41], while match filtering and end-to-end correspondence reasoning have progressed through SuperGlue, NCNet, Patch2Pix, COTR, LoFTR, ASpanFormer, LightGlue, DKM, RoMa, Efficient LoFTR, OmniGlue, MINIMA, and XoFTR [9, 42, 43, 44, 10, 11, 12, 13, 14, 45, 46, 16, 15]. In particular, these methods are better at recovering useful correspondences in low-texture regions, repetitive structures, or substantial appearance shifts. Broad reviews of this transition from classical pipelines to trainable matching, together with practical benchmarking studies, provide additional context [47, 48].

Our work builds directly on this progress by treating modern matchers as high-capacity evidence generators. The empirical picture in UAV-TAlign-12K is that strong off-the-shelf multimodal matchers and infrared-visible classical baselines already provide very high pairwise availability. This moves the central question from correspondence generation alone to the conversion of frame-level estimates into stable and reliable scene-level alignment under realistic UAV operating conditions.

2.3. Visible-Thermal Registration and Evaluation

Visible-thermal registration has been studied under a range of assumptions, from same-view or near-same-view alignment to broader multimodal correspondence learning. Radiation-variation insensitive matching methods such as RIFT and deep alignment or stitching formulations such as CLKN and UDIS indicate the breadth of existing registration formulations [49, 50, 51]. Many existing settings emphasize relatively controlled viewpoint changes or pairwise registration outcomes. By contrast, practical UAV infrared-visible data frequently combine cross-FOV capture, cross-resolution sensing, thermal rendering variation, and low-light conditions. These factors make scene-level reliability a more central concern than in many standard visible-thermal evaluation setups.

The most closely related UAV alignment benchmark direction is represented by synthetic transform-supervised visible-infrared alignment datasets and trainable baselines, such as the recent TGRS UAVMatch benchmark and transformer-based alignment network [17]. That line provides controlled affine or homography supervision, train/test splits at large scale, and geometric metrics such as RMSE, AEE, and PCK. UAV-TAlign-12K is designed for the complementary real-data setting: it fixes an integrity-checked captured cross-FOV evaluation split, evaluates off-the-shelf infrared-visible evidence generators, and studies how frame-level homographies become retained scene-level alignment products under a label-free reliability protocol.

UAV-TAlign-12K extends this evaluation landscape by packaging real cross-FOV infrared-visible scenes together with scalable label-free diagnostics. The protocol measures scene consistency and relative alignment quality, allowing the study to focus on reliability-controlled scene-level alignment under practical UAV acquisition conditions rather than isolated pairwise correspondence counts.

2.4. UAV Multimodal Datasets and the Remaining Gap

UAV multimodal datasets have supported progress in detection, tracking, segmentation, and broader cross-modal perception, spanning aligned pedestrian and automotive benchmarks as well as visible-thermal detection and tracking benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV [1, 2, 3, 4, 5, 6, 7, 8]. These resources have primarily advanced downstream perception tasks, while scene-level infrared-visible alignment places additional emphasis on reliability-aware geometric decision making. In particular, a practical registration pipeline must decide not only how to align frames, but also when the scene-level evidence is strong enough to trust the resulting transform.

This is the gap our paper targets. Complementary to new matcher architectures, we focus on a practical scene-level registration setting for real UAV infrared-visible imagery. Our contribution lies in a reliability-oriented alignment framework, a public 12K-scale collection, and a label-free protocol that make scene-level infrared-visible reliability directly studyable.

3. UAV-TAlign-12K and Evaluation Protocol

3.1. Dataset Scope

We package UAV-TAlign-12K as a public collection for real UAV infrared-visible registration. The collection contains 6,039 candidate RGB-infrared pairs, corresponding to 12,078 images across 15 UAV scenes. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs and 12,074 images. Each sample is organized as an RGB image paired with a thermal image captured from the same scene context, while the collection spans practical UAV conditions that go beyond standard near-same-view multimodal matching. In scope and modality, it is closer to recent visible-thermal benchmarks such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV than to classical same-sensor image matching setups [1, 2, 3, 4, 5, 6, 7, 8].

All main journal-scale experiments are tied to the official manifest with canonical SHA256 prefix `a092f7ad...bb202c6c`. This manifest-based entry point fixes the evaluated split and prevents direct folder traversal from silently changing the reported results. We retain UAV-TAlign-1K-Lite as a development and ablation subset, while UAV-TAlign-12K is the main benchmark-scale evaluation collection for this article.

Table 1: Positioning of UAV-TAlign-12K relative to representative infrared-visible and UAV multimodal resources. UAVMatch denotes the recent synthetic transform-supervised UAV visible-infrared alignment benchmark.

Resource	Data basis	Alignment signal	Scale	Primary evaluation focus
KAIST / FLIR / LLVIP / M3FD	Ground or mixed RGB-infrared data	Paired or aligned imagery	Dataset-specific	Detection, fusion, low-light perception
DroneVehicle / VTUAV / LasHeR / Anti-UAV	UAV or tracking-centric RGB-infrared data	Task annotations	Dataset-specific	Detection or tracking under multimodal sensing
UAVMatch / TGRS [17]	Synthetic transforms from UAV multimodal data	Affine / homography supervision	81K train / 15K test pairs	Supervised transform regression and geometric metrics
UAV-TAlign-12K	Real captured cross-FOV UAV infrared-visible data	Manifest-fixed scene reliability protocol	6,039 candidate / 6,037 valid pairs	Pairwise evidence and scene-level retained alignment

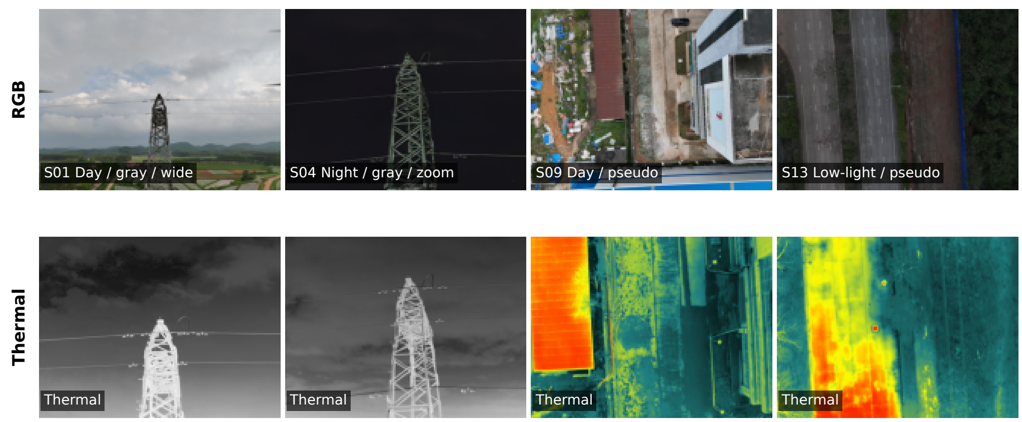

Figure 1: Representative UAV-TAlign-12K RGB-infrared pairs spanning day, night, low-light, grayscale-thermal, pseudocolor-thermal, wide-view, and zoom-view conditions.

At the scene level, UAV-TAlign-12K preserves the factors that make reliability assessment meaningful: day, night, and low-light capture; grayscale and pseudocolor thermal rendering; and both wide-view and zoom-view scene families. The packaged formal split includes 7 day scenes, 5 night scenes, and 3 low-light scenes; 8 grayscale-thermal scenes and 7 pseudocolor-thermal scenes; and paired wide/zoom controls for the substation scene family.

3.2. Dataset Organization

Each scene is stored under its own scene directory, and paired samples are formed by matching file stems between `rgb/` and `thermal/` subdirectories. The dataset loader constructs scene pairs from common RGB and thermal filenames, while scene-level metadata are loaded from the official manifest. For each pair, the loader stores scene identity, pair identity, light condition, thermal rendering, view type, and scene label.

This organization supports both pairwise baseline evaluation and scene-level pipeline evaluation under the same manifest definition, while also enabling condition-based breakdowns from a unified scene manifest.

3.3. Complementary Evaluation Units

The protocol uses two complementary evaluation units. First, off-the-shelf baselines are evaluated at the pair level. Each of the 6,037 integrity-checked RGB-infrared pairs is processed independently, and the resulting statistics describe whether a method can produce usable pairwise homography evidence under a fixed off-the-shelf setting.

Second, UAV-TAlign is evaluated at the scene level. Multiple frame-level estimates are aggregated into a scene-level transform, and the final decision is made by canonical scene acceptance rather than by pairwise usability alone. The pairwise and scene-level tables therefore provide two complementary views of practical registration behavior: correspondence availability and retained scene-level alignment reliability.

3.4. Label-Free QA Proxies

The protocol relies on label-free relative QA signals designed for scalable scene-level reliability analysis. The strongest stored proxies compare the optimized transform against a baseline initialization and summarize whether scene-level rectification improves cross-modal agreement. The proxy family includes edge-overlap improvement, gradient-NCC improvement, the ratio of frames with improved gradient agreement, severe baseline-relative outlier count and ratio, accepted-frame ratio, and homography-dispersion and robust-rejection diagnostics.

These proxies provide a practical scene-consistency signal for comparing methods and for deciding whether a scene-level transform is reliable enough to retain. They provide scalable scene-consistency diagnostics for reliability-controlled evaluation and can be combined with task-specific annotation protocols when such labels are available, in the spirit of practical matching-oriented evaluation and benchmarking discussed in prior image-matching literature [47, 48].

4. Method

4.1. Problem Formulation

We study infrared-visible registration as a scene-level alignment task. For each scene, the input is a set of paired RGB and thermal UAV frames $S = \{(I_i^R, I_i^T)\}_{i=1}^N$, where viewpoint variation, cross-resolution sensing, thermal appearance changes, and day/night conditions make individual frame

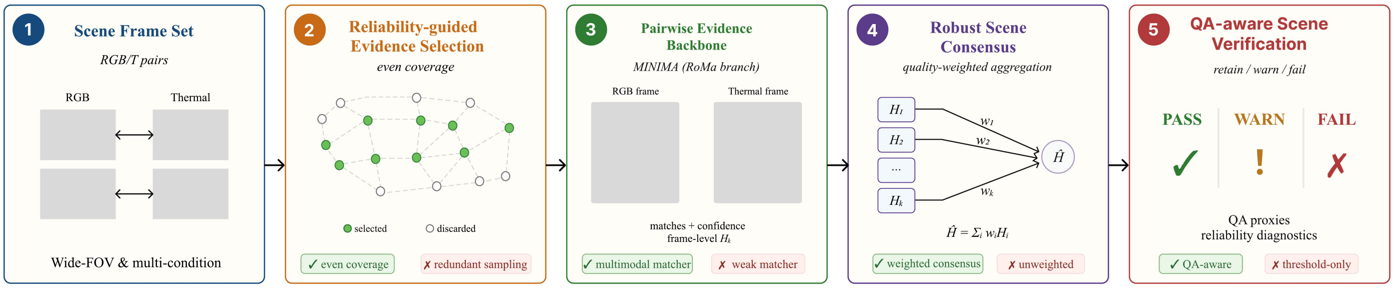

Figure 2: Overview of UAV-TAlign. The framework builds on strong pairwise matching, but makes the final decision at the scene level through reliability-guided evidence selection, robust scene consensus, and QA-aware verification.

estimates uneven in quality. The goal is to recover a thermal-to-RGB homography that remains reliable at the scene level, with the canonical output retained when the evidence satisfies the reliability criterion.

This formulation turns registration from a single best-pair problem into a multi-frame reliability optimization problem. In our setting, many scenes already contain usable pairwise correspondences, but the central challenge is to convert uneven frame-level estimates into a transform that is geometrically stable, internally consistent, and acceptable under a reliability-aware criterion. For a selected frame pair, the thermal-to-RGB evidence is represented as a homography $H_i \in \mathbb{R}^{3\times3}$ acting on homogeneous image coordinates,

$$\tilde{\mathbf{p}}_{ij}^{R} \sim H_i \tilde{\mathbf{p}}_{ij}^{T}, \qquad H_i = \arg\min_{H} \sum_{j\in\mathcal{I}_i} \rho\left(\left\| \pi(H\tilde{\mathbf{p}}_{ij}^{T}) - \mathbf{p}_{ij}^{R} \right\|_2^2 \right), \tag{1}$$

where $\mathcal{I}_i$ denotes the inlier set selected by robust fitting, $\pi(\cdot)$ converts a homogeneous coordinate to image coordinates, and $\rho(\cdot)$ is the robust estimator loss. The scene-level problem is then to estimate a consensus transform $\bar{H}_S$ from uneven evidence $\{H_i\}$ and decide whether it should be retained as an alignment product.

4.2. Pairwise Evidence Backbone

UAV-TAlign uses a strong off-the-shelf multimodal matcher as its pairwise evidence generator. Our implementation uses the MINIMA family with the RoMa branch as the default backend, while the rest of the pipeline remains agnostic to the exact pairwise matcher [14, 16].

4.3. Reliability-Guided Evidence Selection

At the scene level, UAV-TAlign prioritizes the most informative frame evidence before expanding the support set. It uses a progressive evidence schedule that expands with scene size and can cover all frames when needed. This reliability-guided policy allows the framework to stop when a compact subset already yields a stable consensus, while still escalating to broader evidence when the scene benefits from more support.

Within each stage, representative frames are selected with deterministic evenly distributed sampling by default. This default policy is used for reproducibility and stable scene coverage. A random-selection variant is reserved for sensitivity analysis and is reported separately.

4.4. Quality-Weighted Frame Geometry

For each selected frame pair, the matcher produces cross-modal correspondences that are filtered into a frame-level homography estimate. UAV-TAlign evaluates each frame using standard geometric statistics, including the number of matches, the number of inliers, inlier ratio, reprojection error, and spatial coverage. A frame enters the global candidate pool only when it passes a geometry-quality gate on these statistics. The overall geometric treatment remains rooted in standard homography estimation and robust fitting practice [25, 24, 26, 27].

Beyond a binary accept/reject decision, the pipeline assigns each accepted frame a quality score so that better-supported homographies receive more influence during scene aggregation. The score is a weighted combination of inlier ratio, spatial coverage, reprojection quality, matcher confidence, and inlier count. This keeps the scene-level estimate from being dominated by low-support frames that only marginally satisfy the frame-level gate. In normalized form, the frame quality weight is written as

$$q_i = \alpha_r\phi(r_i)+\alpha_c\phi(c_i)+\alpha_e\phi(e_i)+\alpha_m\phi(m_i)+\alpha_n\phi(n_i), \qquad \sum_k \alpha_k = 1,\ \alpha_k \geq 0, \tag{2}$$

where r_i is inlier ratio, c_i is spatial coverage, e_i is reprojection quality, m_i is matcher confidence, n_i is inlier support, and $\phi(\cdot)$ maps each diagnostic into a consistent reliability direction. The weights are fixed by the evaluation protocol and are not learned from UAV-TAlign-12K.

4.5. Robust Scene Consensus

Once enough frame-level homographies have been accepted, UAV-TAlign forms a robust scene consensus. Homographies are mapped into parameter space, centered with a median reference, and filtered by a median-absolute-deviation rule to suppress unstable estimates before weighted averaging. The retained homographies are then combined using the frame quality scores defined above. Let $\mathbf{h}_i$ be the scale-normalized vector form of H_i. The consensus set is

$$\mathcal{C}_S = \left\{ i \in \mathcal{A}_S : \left\| \mathbf{h}_i - \mathrm{median}_{j \in \mathcal{A}_S}(\mathbf{h}_j) \right\|_1 \leq \tau_{\mathrm{mad}} \ \mathrm{MAD}(\{\mathbf{h}_j\}_{j \in \mathcal{A}_S}) \right\}, \quad (3)$$

where $\mathcal{A}_S$ is the accepted-frame set before robust consensus. The scene transform is then obtained by quality-weighted aggregation,

$$\bar{\mathbf{h}}_S = \frac{\sum_{i \in \mathcal{C}_S} q_i \mathbf{h}_i}{\sum_{i \in \mathcal{C}_S} q_i}, \qquad \bar{H}_S = \mathrm{mat}(\bar{\mathbf{h}}_S). \quad (4)$$

The pipeline records stability diagnostics around the aggregate transform, including homography dispersion relative to the aggregate and the ratio of robustly rejected frame-level estimates. These diagnostics connect consensus formation with the final QA-aware decision.

4.6. QA-Aware Scene Verification

The final decision integrates match availability with scene-level reliability evidence. UAV-TAlign evaluates the optimized scene-level transform against a label-free QA summary computed on representative frames. The QA summary uses cross-modal proxy signals such as edge-overlap improvement, gradient-NCC improvement, the ratio of improved frames, and the count of severe relative outliers. It also retains scene-level match and stability statistics such as accepted-frame ratio, mean inlier ratio, mean reprojection error, spatial coverage, dispersion relative to the aggregate transform, and the robust reject ratio.

The canonical decision uses a baseline-aware verification rule. A scene is retained when accepted-frame statistics remain strong, the aggregated homography is stable, severe baseline-relative outliers remain controlled, and the transform improves gradient-based alignment beyond configured floors. This final verification step gives UAV-TAlign its scene-level character: the framework prioritizes retained scene-level usability over unconditional scene acceptance. For operating-profile visualization, the same diagnostics are summarized by a non-trained reliability score,

$$
\begin{aligned}
R(S) = {} & \beta_a A_S + \beta_o(1 - O_S) + \beta_d(1 - D_S) \\
& + \beta_e \Delta E_S + \beta_g \Delta G_S + \beta_\gamma \Gamma_S, \qquad \sum_k \beta_k = 1.
\end{aligned} \tag{5}
$$

where A_S is accepted-frame support, O_S is severe baseline-relative outlier ratio, D_S is homography dispersion, ΔE_S and ΔG_S are edge and gradient improvement proxies, and Γ_S collects geometry diagnostics. The strict canonical gate remains a thresholded decision on the underlying diagnostics,

$$
y_S = \mathbb{1}\left[A_S \geq a_{\min},\ O_S \leq o_{\max},\ D_S \leq d_{\max}, \Delta G_S \geq g_{\min},\ \Gamma_S \in \mathcal{G}\right]. \tag{6}
$$

Thus $R(S)$ ranks scenes for risk-coverage analysis, while y_S defines the retained canonical alignment products reported in the main result.

5. Experimental Setup

5.1. Baselines

We evaluate both classical and learning-based baselines under an explicit off-the-shelf setting. All learning-based methods use public pretrained weights or public pretrained model interfaces and are evaluated without training or adaptation on UAV-TAlign-12K. This choice keeps the comparison focused on practical deployability rather than dataset-specific adaptation. Unified inference wrappers standardize model loading, image I/O, homography fitting, and output schemas while preserving the public pretrained baseline parameters and settings.

The baseline suite includes a domain-specific infrared-visible classical method and modern dense/learned matchers. RIFT2 is evaluated through a Python implementation under a resize-aware setting. The modern pairwise baselines are Kornia LoFTR (pretrained="outdoor"), RoMa outdoor full, XoFTR-640, and raw MINIMA. Raw MINIMA uses the MINIMA RoMa branch as a direct pairwise baseline; reliability-guided evidence selection, robust scene consensus, and QA-aware verification are activated in the UAV-TAlign full framework [49, 10, 14, 15, 16].

5.2. Shared Geometric Fitting

For the pairwise methods, the common downstream step is homography fitting from the predicted correspondences. After each matcher returns point pairs and optional confidence values, the shared evaluator estimates a homography with a robust backend and then records a unified outcome category. This unified post-processing is important because it makes the pairwise baseline table comparable at the level of usable registration outcomes rather than raw matcher outputs alone.

The common evaluator uses a robust homography backend across the modern pairwise baselines:

- robust fitting method: USAC-MAGSAC;
- reprojection threshold: 3.0;
- confidence: 0.999;
- maximum iterations: 10000;
- spatial coverage grid: 4×4.

RIFT2 is reported as *RIFT2 (Python, resize-aware)*. Its disclosed configuration is `max_dim=1200`, `npt=2000`, `patch_size=64`, Lowe ratio 0.95, and OpenCV USAC-MAGSAC for downstream homography estimation. We report it as a fixed Python resize-aware deployment setting for the UAV-TAlign-12K evaluation.

5.3. Main-Table Baseline Choices

The main-table baseline choices are as follows. Raw MINIMA uses the MINIMA RoMa backend with the `minima_roma` checkpoint family and the large/full RoMa branch. RoMa uses the public outdoor full model. XoFTR uses the 640-resolution visible-thermal checkpoint family as the default main-table setting. Kornia LoFTR (pretrained="outdoor") uses the public pretrained outdoor model from Kornia.

5.4. Evaluation Details

The LoFTR baseline is evaluated under a resize-aware inference setting. The longest image dimension used for matching is limited to 1600, and automatic mixed precision is disabled. This deployment setting preserves the

Table 2: Pairwise infrared-visible homography evidence on UAV-TAlign-12K. Metrics are complementary diagnostics under the evaluated implementation and deployment settings, and should not be read as a single absolute ranking. LoFTR and XoFTR non-ok cases correspond to insufficient matches, no matches, or homography fitting failures.

Method	Category	OK / N ↑	H (%) ↑	Inlier Ratio ↑	Coverage ↑	Median Reproj. ↓	Runtime / Pair (s) ↓
SIFT + RANSAC	Classical keypoint	5257/6037	87.08	0.242	0.627	0.000	9.008
AKAZE + RANSAC	Classical keypoint	5692/6037	94.29	0.160	0.730	0.105	1.511
RIFT2 (Python, resize-aware)	Infrared-visible classical	6037/6037	**100.00**	0.086	0.354	0.121	13.980
raw MINIMA	Modern dense / learned	6037/6037	**100.00**	0.319	1.000	1.775	1.905
RoMa outdoor	Modern dense / learned	6037/6037	**100.00**	0.328	0.997	1.739	0.844
Kornia LoFTR outdoor	Modern dense / learned	6009/6037	99.54	0.099	0.922	1.405	0.288
XoFTR-640	Modern dense / learned	5989/6037	99.20	0.089	0.949	1.732	0.615

public pretrained model while supporting consistent full-subset evaluation within the fixed deployment budget.

Journal-scale experiments were conducted in isolated Python environments on an NVIDIA RTX 4090 workstation. The main environment used Python 3.10 with PyTorch 2.0.1, torchvision 0.15.2, CUDA 11.8, OpenCV 4.11.0, and Kornia 0.6.11. RIFT2 ran separately under Python 3.11 with OpenCV 4.10.0 and NumPy 1.26.4. All runs used the fixed official manifest, fixed public weights or model interfaces, isolated output roots, and read-only raw dataset directories.

6. Results

6.1. Pairwise Infrared-Visible Homography Evidence

Pairwise infrared-visible homography evidence is broadly available on UAV-TAlign-12K. Table 2 reports the full 6,037-pair official evaluation split. Classical keypoint baselines provide useful but less complete pairwise evidence: SIFT and AKAZE reach 87.08% and 94.29% homography availability, respectively. This provides a classical keypoint reference, while RIFT2 and modern dense and learned matchers further improve pairwise availability under infrared-visible appearance gaps. RIFT2 adds an infrared-visible classical reference point and completes all evaluated pairs. Raw MINIMA and RoMa also produce homographies for all pairs, while LoFTR and XoFTR remain near-saturated under the same manifest-based evaluation. This result sharpens the role of UAV-TAlign: the framework targets the decision layer that remains after strong pairwise homography evidence is available.

RIFT2 provides stable infrared-visible pairwise homographies with low fitted reprojection error, while modern dense and learned matchers provide

Table 3: Strict canonical scene-level result for UAV-TAlign on UAV-TAlign-12K. Arrows indicate the preferred direction.

Method	Eval Unit	H Available ↑	Retained Scenes ↑	Retention Rate (%) ↑	Accepted Frames	Accepted Ratio (%) ↑
UAV-TAlign	15 scenes	15/15	9/15	60.0	2473/3847	64.3

Table 4: Scene-level reliability breakdown by light condition on UAV-TAlign-12K.

Condition	Scenes	Retained ↑	Retention (%) ↑	Accepted / Attempted ↑	Accepted Ratio (%) ↑
Day	7	**5**	**71.4**	**1650/2203**	**74.9**
Night	5	3	60.0	361/890	40.6
Low-light	3	1	33.3	462/754	61.3

broader spatial support and stronger coverage. These secondary fitting statistics are therefore best read jointly.

6.2. Scene-Level Performance of UAV-TAlign

High pairwise usability does not automatically imply broad scene retention, because the canonical decision answers a different operating question. UAV-TAlign produces scene-level homographies for all 15 scenes and retains 9 scenes under the strict canonical reliability gate. This operating point emphasizes high-confidence alignment products and shows how robust consensus and verification turn abundant frame-level matches into selected scene-level transforms.

6.3. Condition Breakdown

Retained alignments show meaningful condition structure across illumination and thermal rendering. Day, night, and low-light groups all contribute retained products, while their accepted-frame profiles differ substantially. These patterns support the view that the central challenge is scene-level decision quality rather than pairwise matcher availability alone.

6.4. Reliability-Coverage Operating Profile

The canonical QA gate defines retained alignment products, while the reliability score is used for ranking, threshold sweep, and operating-profile visualization. The score-ranked top-K subsets and the canonical retained-scene set are related but not identical. Figure 4 therefore marks the strict canonical 9/15 operating point explicitly while plotting the score-ranked reliability–coverage curve.

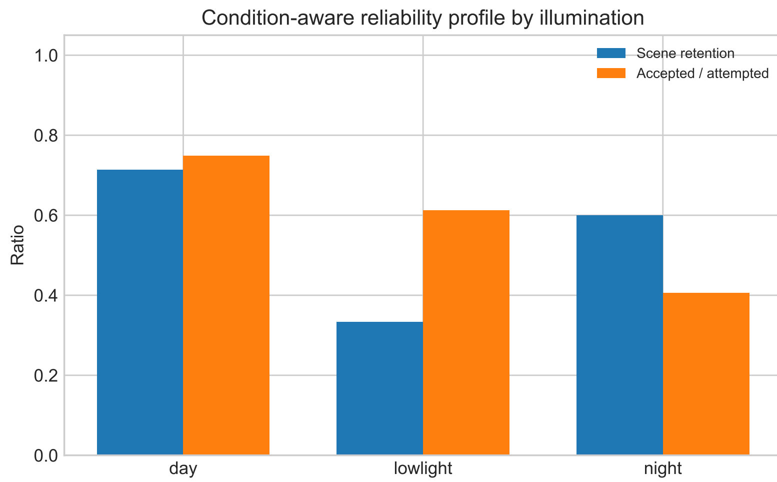

Figure 3: Condition-aware reliability profile by illumination. Scene retention and accepted-frame support summarize complementary aspects of the retained alignment products.

At the canonical point, UAV-TAlign retains 9 scenes, covers 60.0% of scenes and 74.4% of evaluated pairs, and keeps the mean severe-outlier ratio at 0.009. The score-ranked curve exposes a smooth operating profile from conservative high-confidence retention to broader inclusive coverage. We keep the canonical gate fixed and use this curve to visualize the reliability–coverage trade-off.

6.5. Cumulative Ablation

The strongest module-level gain is produced by robust scene consensus, while deterministic selection is best understood as a reproducible operating policy. Adding robust scene consensus substantially improves edge and gradient alignment proxies while reducing severe outliers. Adding QA-aware verification then converts accepted-frame evidence into a reliability-controlled scene-level criterion, matching the 1K-Lite ablation slice at 5/8 retained scenes.

The completed random-selection supplement further shows a 4/8 to 7/8 spread across seeds on the same fixed subset. Deterministic-even selection therefore serves as a reproducible default operating policy, while robust scene consensus and QA-aware verification remain the primary empirical drivers.

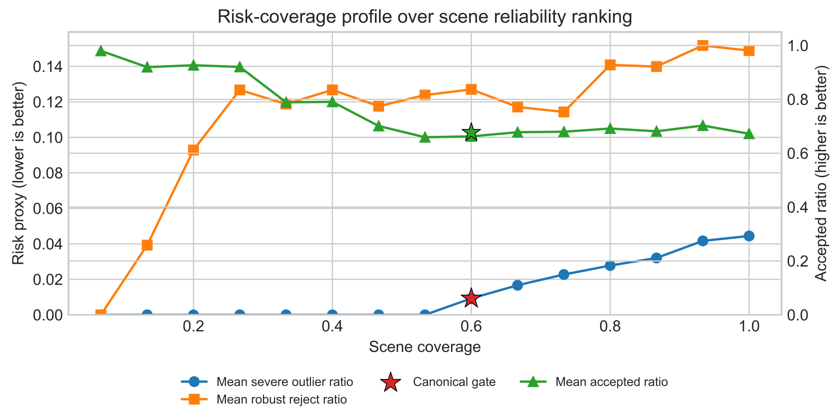

Figure 4: Risk-coverage operating profile on UAV-TAlign-12K. The curves are generated by ranking scenes with the non-trained reliability score and progressively increasing retained coverage. The star marks the strict canonical QA-gate operating point (9/15 retained scenes). The score-ranked top-K operating points visualize the reliability–coverage profile and do not replace the canonical decision rule.

Table 5: Cumulative module ablation on the fixed 8-scene subset. Bold values mark the strongest observed proxy values within each column; arrows indicate preferred direction.

Variant	Evidence	Consensus	Verification	Retained/N ↑	Acc. ↑	ΔEdge ↑	ΔGrad ↑	Severe ↓
Evidence selection only	✓	–	–	8/8	77.4	0.124	0.126	10.4
+ Robust scene consensus	✓	✓	–	8/8	77.4	0.145	0.148	**4.2**
+ QA-aware verification	✓	✓	✓	5/8	77.2	**0.147**	**0.149**	**4.2**
UAV-TAlign full (8-scene slice)	✓	✓	✓	5/8	**80.4**	**0.147**	**0.149**	**4.2**

7. Discussion

7.1. Why Scene-Level Reliability Becomes Central

The results position UAV-TAlign as a scene-level reliability framework for UAV infrared-visible alignment. Modern pairwise matchers already provide abundant frame-level correspondences on UAV-TAlign-12K, which makes scene-level decision quality the central operating question. UAV-TAlign addresses this layer directly by elevating strong pairwise correspondence models into reliability-controlled scene-level alignment products.

The protocol evidence supports this reliability framing. Retained and QA-filtered scenes show structured differences in accepted ratio, severe outlier behavior, and baseline-relative QA signals, indicating that the canonical

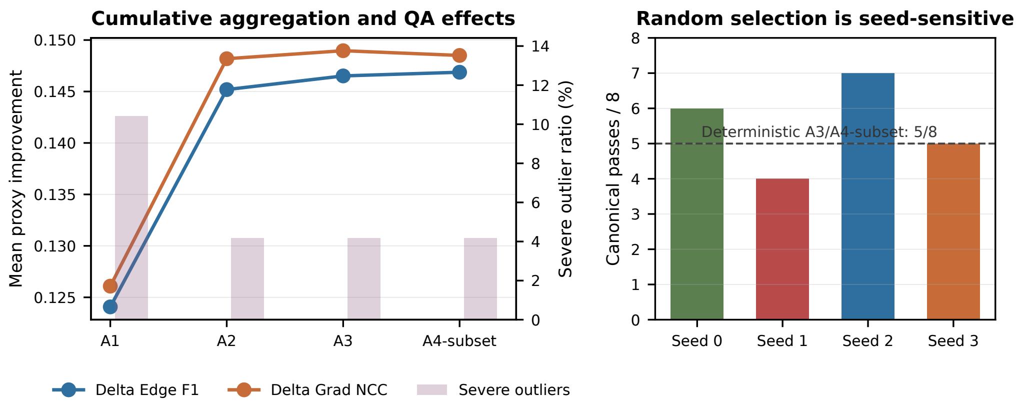

Figure 5: Cumulative ablation and random-selection supplement. Robust scene consensus improves proxy quality and reduces severe outliers, while the random-selection variant shows non-trivial seed sensitivity on the same fixed 8-scene subset.

criterion captures scene-level reliability beyond pairwise matcher availability. This gives the evaluation a clear diagnostic role: it identifies when abundant pairwise evidence becomes a trustworthy scene-level transform. The canonical scene criterion is designed for conservative retention of scene transforms, not for maximizing unconditional scene acceptance.

The completed multi-seed random-selection supplement further strengthens the design rationale for deterministic evidence selection: it provides a stable operating policy under the same fixed scene subset, while robust scene consensus and QA-aware verification deliver the strongest module-level gains.

7.2. Scope and Validity

UAV-TAlign targets real UAV scenes where a dominant planar or far-field transform is a useful operating model. Within this scope, public collection packaging, label-free diagnostics, and scene-level QA provide a benchmark-scale experimental foundation for studying infrared-visible alignment beyond isolated pairwise matching. This positioning is deliberately complementary to matcher-centric research and synthetic transform-supervised UAV alignment benchmarks: as correspondence models and trainable transform predictors continue to improve, reliability control becomes increasingly important for deployable infrared-visible systems operating on captured cross-FOV scenes.

The protocol is designed for scalable reliability analysis. Its QA proxies operate as scene-consistency diagnostics for retained alignment products, while the risk-coverage profile shows how operating coverage changes under a fixed reliability-ranking view. The homography model is most appropriate when a dominant planar or far-field alignment is meaningful, and the UAV-TAlign-12K collection focuses on this practical operating regime for UAV infrared-visible alignment.

8. Conclusion

This work reframes UAV infrared-visible registration from pairwise match availability to scene-level alignment reliability. UAV-TAlign turns strong off-the-shelf multimodal matching into retained scene-level decisions through reliability-guided evidence selection, robust scene consensus, and QA-aware verification. Across formal evaluation, protocol-consistency analysis, and cumulative module ablations, the evidence shows that the decisive step is transforming abundant frame-level matches into trustworthy scene-level alignment products.

Together with UAV-TAlign-12K and the accompanying label-free evaluation setup, UAV-TAlign provides a practical and reproducible foundation for studying when scene-level alignment is reliable enough to retain in real UAV infrared-visible imagery.

Data Availability

UAV-TAlign-12K, the official evaluation manifest, metadata, integrity report, and evaluation scripts have been prepared for public release. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs from a 6,039-pair UAV infrared-visible alignment collection. The dataset and code will be released through a public repository, and review access can be provided during peer review. The official manifest hash is provided for reproducible evaluation.

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